

Health Chatbot System

Comprehensive Technical Answers for Company Review

ULSS 9 Scaligera Chatbot Team

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1 Executive Summary

This document provides comprehensive technical answers regarding the Health Chatbot system (ULSS 9 Scaligera) for company evaluation. The system is built on Google Cloud Platform using Gemini AI with a focus on **privacy, security, and enterprise-grade data protection**.

Key Highlights

- **Privacy-First Architecture:** Customer data is NOT used to train public Gemini models
- **Secure RAG Implementation:** Gemini File Search with encrypted, isolated storage
- **Enterprise Controls:** VPC-SC, CMEK, data residency, and compliance certifications
- **Scalable Infrastructure:** Auto-scaling with support for thousands of concurrent users
- **Transparent Pricing:** Pay-per-use model with predictable costs

2 Architecture & Privacy Protection

2.1 System Architecture Overview

[CURRENT IMPLEMENTATION]

The Health Chatbot system follows a three-tier architecture designed for maximum security and privacy:

1. **Frontend Layer:** Next.js application (user interface)
2. **Backend Layer:** FastAPI application managing business logic and API orchestration
3. **AI/RAG Layer:** Google Cloud Gemini API with File Search Stores

Data Flow

User Query → Frontend → Backend API → Gemini API (with File Search) → Response

2.2 How Documents Remain Private

2.2.1 No Public Model Training

Critical Guarantee: When using the **paid tier** of Google Cloud's Gemini API, customer data (prompts, responses, and uploaded documents) is **explicitly NOT used** to train or improve Google's foundational models.

*"Google Cloud does not use customer data, including prompts, responses, and fine-tuning data, to train or fine-tune any of its AI/ML models without the customer's prior permission or instruction."*¹

What this means for sensitive health data:

¹<https://cloud.google.com/vertex-ai/generative-ai/docs/learn/responsible-ai>

- Medical statistics, patient information, and internal documents remain completely private
- No data leakage to public models or other customers
- Data is processed only for your specific queries and remains isolated

2.2.2 Data Encryption & Isolation

All uploaded documents and queries are protected by multiple layers of security:

Security Layer	Protection
Encryption in Transit	TLS 1.3 for all API communications
Encryption at Rest	AES-256 encryption for stored embeddings
Data Isolation	Per-customer File Search Stores
Access Control	API key-based authentication
Network Security	VPC Service Controls (optional)

Table 1: Multi-Layer Security Architecture

Reference: <https://cloud.google.com/security/encryption>

2.3 Enterprise Privacy Controls

For maximum privacy protection, the following Google Cloud features can be enabled:

2.3.1 VPC Service Controls (VPC-SC)

Creates a security perimeter around your resources to prevent data exfiltration:

- Restricts access to Google Cloud APIs from outside defined perimeters
- Prevents accidental or malicious data leakage
- Particularly important for sensitive healthcare data

Reference: <https://cloud.google.com/vpc-service-controls/docs/overview>

2.3.2 Customer-Managed Encryption Keys (CMEK)

Allows you to control your own encryption keys:

- You manage the cryptographic keys used to encrypt data
- Can revoke access at any time by revoking keys
- Supports Hardware Security Modules (HSM) for maximum security

Reference: <https://cloud.google.com/vertex-ai/docs/general/cmek>

2.3.3 Data Residency Controls

Ensures data remains in specific geographic regions:

- Configure data processing to EU or US regions only
- Complies with GDPR and regional data sovereignty requirements
- Important for Italian/European healthcare regulations

Reference: <https://cloud.google.com/vertex-ai/docs/general/locations>

2.4 Compliance Certifications

Google Cloud Vertex AI (including Gemini API) maintains rigorous compliance certifications:

- **HIPAA Compliance:** For Protected Health Information (with BAA)
- **ISO 27001, 27017, 27018, 27701:** Information security standards
- **SOC 1, 2, 3:** Service organization controls
- **GDPR Compliant:** European data protection regulations

Reference: <https://cloud.google.com/security/compliance>

3 RAG Implementation & Alternatives

3.1 Current Implementation: Gemini File Search

[WHAT WE ARE USING NOW]

Our chatbot uses **Gemini File Search Stores**, a managed RAG (Retrieval Augmented Generation) solution provided by Google Cloud.

Why this choice: Native integration, enterprise security, zero DevOps overhead.

3.1.1 How It Works

1. **Document Upload:** PDF, DOCX, TXT files uploaded via Admin API
2. **Processing:** Documents are chunked and converted to semantic embeddings
3. **Storage:** Embeddings stored in File Search database per domain
4. **Retrieval:** User queries trigger semantic search across relevant stores
5. **Generation:** Gemini model generates contextual answers using retrieved chunks

File Search Stores configured in production:

- **ulss9-general_info:** General healthcare information
- **ulss9-hours:** Operating hours, schedules
- **ulss9-locations:** Facility addresses, maps
- **ulss9-services:** Medical services, procedures

3.1.2 Data Lifecycle

- **Raw Files:** Deleted after 48 hours (converted to embeddings)
- **Embeddings:** Persist indefinitely in encrypted storage
- **Infrastructure:** Fully managed by Google (no server maintenance)

Reference: <https://ai.google.dev/gemini-api/docs/file-search>

Criterion	Advantage
Integration	Native integration with Gemini API—no separate vector DB setup
Security	Inherits all Google Cloud security features (encryption, VPC-SC, CMEK)
Managed Infrastructure	Zero DevOps overhead; Google handles scaling, backups, updates
Cost Efficiency	No separate vector database fees; bundled with Gemini API usage
Privacy	Data stays within Google Cloud ecosystem with isolation guarantees
Ease of Use	Simple API for document upload and management

Table 2: Gemini File Search Advantages

3.2 Why We Chose Gemini File Search

3.3 Alternative RAG Solutions Considered

[ALTERNATIVES - NOT CURRENTLY USED]

We evaluated the following alternatives before selecting Gemini File Search, but they are **NOT part of our current implementation**:

3.3.1 Vertex AI Search (formerly Enterprise Search)

Description: Google’s enterprise-grade search solution with advanced RAG capabilities.

- **Pros:** Enhanced enterprise features, multi-corpus segmentation, hybrid search
- **Cons:** Higher complexity, premium pricing, overkill for our current scale
- **Use Case:** Best for large enterprises with complex search requirements (1M+ documents)

Reference: <https://cloud.google.com/enterprise-search>

3.3.2 Pinecone

Description: Fully managed, proprietary vector database (SaaS).

- **Pros:** Excellent performance, zero DevOps, strong SLAs
- **Cons:** Premium pricing, vendor lock-in, data stored outside Google Cloud
- **Why Not Chosen:** Additional vendor dependency; data would leave GCP ecosystem

Reference: <https://www.pinecone.io>

3.3.3 Weaviate

Description: Open-source vector database with cloud and self-hosted options.

- **Pros:** Hybrid search, flexible deployment, GraphQL interface, lower cost
- **Cons:** Requires infrastructure management if self-hosted
- **Why Not Chosen:** Added operational complexity; requires separate deployment and maintenance

Reference: <https://weaviate.io>

3.3.4 ChromaDB

Description: Lightweight, Python-native, open-source vector database.

- **Pros:** Simple setup, great for prototyping, free and open-source
- **Cons:** Limited scalability, not production-grade for high-traffic systems
- **Why Not Chosen:** Not suitable for enterprise healthcare application requirements

Reference: <https://www.trychroma.com>

3.4 RAG Type: Vector-Based Semantic Search

Our implementation uses **dense vector embeddings** for semantic retrieval:

- **Type:** Embedding-based RAG (not keyword-based or graph-based)
- **Model:** Google's text embedding models (managed by File Search)
- **Similarity:** Cosine similarity for finding relevant document chunks
- **Reranking:** Automatic relevance scoring by Gemini

4 Data Training & Retention

4.1 Understanding "Training" in RAG Context

It's crucial to distinguish between two types of "training":

Type	Explanation
RAG Document Indexing	Converting documents to embeddings for retrieval (happens in our system)
LLM Model Training	Training the foundational Gemini model itself (does NOT happen with our data)

4.2 What Happens to Your Documents

1. **Upload:** Documents uploaded via Admin API to File Search Store
2. **Processing:**
 - Documents chunked into smaller segments
 - Each chunk converted to vector embeddings (numerical representations)
 - Embeddings stored in encrypted database
3. **Raw File Deletion:** Original files deleted after 48 hours
4. **Embedding Persistence:** Embeddings remain for RAG queries
5. **Query Time:** Embeddings used to find relevant context, NOT to train Gemini

4.3 Data Retention Policy

- **Embeddings:** Persist indefinitely until manually deleted
- **Query Logs:** Can be disabled for zero data retention (optional)
- **Abuse Monitoring:** Limited logging for policy violations (can request exemption)
- **Deletion:** Documents can be removed via API at any time

Reference: <https://cloud.google.com/vertex-ai/generative-ai/docs/learn/responsible-ai>

4.4 Future Training/Fine-Tuning Options

If needed in the future, we could implement:

- **Model Fine-Tuning:** Train a custom Gemini model on domain-specific medical language
 - Data remains in your GCP project
 - Fine-tuned model is exclusive to your organization
 - No data sharing with other customers
- **Prompt Engineering:** Customize system instructions without any training
- **Context Caching:** Cache frequently accessed context for faster responses

Reference: <https://cloud.google.com/vertex-ai/generative-ai/docs/models/tune-models>

5 Cost Analysis & Payment Structure

5.1 Pricing Model Overview

Google Cloud Gemini API uses a **token-based, pay-per-use** pricing model with no upfront costs or commitments.

5.2 Current Model Pricing (Gemini 2.5 Flash)

[CURRENT MODEL IN USE]

Our system uses `gemini-2.5-flash` for optimal cost-performance balance.

Component	Price per 1M Tokens	Notes
Text/Image Input	\$0.15	User queries + retrieved documents
Text Output	\$0.60	Generated responses
Audio Input	\$1.00	If using voice (future feature)
Context Caching	\$0.03 storage + \$0.10 usage	Optional optimization

Table 3: Gemini 2.5 Flash Pricing (as of February 2026)

Reference: <https://ai.google.dev/pricing>

5.3 Token Estimation

Typical usage breakdown:

- **User Query:** 20-50 tokens (average: 30 tokens)
- **Retrieved Context:** 500-2000 tokens from documents (average: 1000 tokens)
- **System Instructions:** 150 tokens (fixed)
- **Response:** 100-500 tokens (average: 250 tokens)

Average per query:

- Input: $\sim 1,180$ tokens
- Output: ~ 250 tokens

5.4 Cost Examples

The following examples include **all costs**: Gemini API + Backend + Frontend + Infrastructure

5.4.1 Example 1: Small Scale (1,000 queries/month)

Breakdown:

- Gemini API Input: $\frac{1,180 \times 1,000}{1,000,000} \times \$0.15 = \$0.18$
- Gemini API Output: $\frac{250 \times 1,000}{1,000,000} \times \$0.60 = \$0.15$
- Backend Hosting (Render/Cloud Run): \$7/month
- Frontend Hosting (Vercel Free): \$0/month
- Database (if needed): \$0 (not required for basic setup)

Total All-Inclusive Cost: \$7.33/month

5.4.2 Example 2: Medium Scale (50,000 queries/month)

Breakdown:

- Gemini API Input: $\frac{1,180 \times 50,000}{1,000,000} \times \$0.15 = \$8.85$
- Gemini API Output: $\frac{250 \times 50,000}{1,000,000} \times \$0.60 = \$7.50$
- Backend Hosting (Render Standard): \$15/month
- Frontend Hosting (Vercel Pro): \$20/month
- Database (PostgreSQL starter): \$7/month

Total All-Inclusive Cost: \$58.35/month

5.4.3 Example 3: Large Scale (500,000 queries/month)

Breakdown:

- Gemini API Input: $\frac{1,180 \times 500,000}{1,000,000} \times \$0.15 = \$88.50$
- Gemini API Output: $\frac{250 \times 500,000}{1,000,000} \times \$0.60 = \$75.00$
- Backend Hosting (Cloud Run/Professional): \$50/month
- Frontend Hosting (Vercel Pro): \$20/month
- Database (PostgreSQL production): \$25/month

Total All-Inclusive Cost: \$258.50/month

Cost Summary

- **Small Clinic (1K queries/month):** \$7.33/month total
- **Medium Hospital (50K queries/month):** \$58.35/month total
- **Large Health System (500K queries/month):** \$258.50/month total

Note: Costs scale primarily with usage. The Gemini API is pay-per-use, while infrastructure costs increase with traffic and reliability requirements.

5.5 Infrastructure Components (Included Above)

The all-inclusive pricing includes the following infrastructure components:

- **Backend Hosting:** FastAPI on Render or Google Cloud Run
 - Small: Render Free/Starter (\$0-7/month)
 - Medium: Render Standard (\$15/month)
 - Large: Cloud Run or dedicated instances (\$50+/month)
- **Frontend Hosting:** Next.js on Vercel
 - Small: Vercel Free tier (\$0/month)
 - Medium/Large: Vercel Pro (\$20/month)
- **Database (optional):** PostgreSQL for user management
 - Not required for basic RAG functionality
 - If needed: \$7-25/month depending on scale
- **Domain & SSL:** \$12-15/year (already included in hosting for most providers)

5.6 Cost Optimization Strategies

1. **Context Caching:** Cache frequently accessed document context (reduces input tokens by 50-70%)
2. **Response Length Limits:** Set max_tokens to prevent unnecessarily long answers
3. **Smart Retrieval:** Limit retrieved chunks to most relevant (reduce context size)
4. **Batch Processing:** Use Batch API for non-real-time queries (if applicable)

5.7 Payment Structure

- **Billing Model:** Monthly subscription-style billing
- **Components:**
 - Gemini API: Pay-per-use via Google Cloud Billing (monthly)
 - Backend: Fixed monthly fee (varies by tier)
 - Frontend: Fixed monthly fee or free tier
- **No Upfront Fees:** All services are pay-as-you-go
- **Cost Monitoring:** Real-time dashboards in Google Cloud Console
- **Billing Alerts:** Set budget alerts to avoid unexpected costs
- **Free Testing:** Gemini API free tier available for development/testing
- **Credits:** Google Cloud credits can be applied to API usage

Typical Monthly Bill Structure:

- Google Cloud (Gemini API): Variable, based on queries
- Render/Vercel (Hosting): Fixed tier pricing
- Total predictable within 10-15% month-to-month for stable usage

Reference: <https://cloud.google.com/billing/docs>

6 Scalability & Concurrent Users

6.1 Rate Limits & Quotas

Google Cloud enforces quotas to ensure fair usage and system stability. These can be increased as needed.

6.1.1 Initial Quotas (New Accounts)

Model	Requests/Minute/Region	Tokens/Minute
gemini-2.5-flash	5	4 million
gemini-2.5-pro	5	4 million
gemini-1.5-flash	15	4 million

Table 4: Default Quotas for Generative AI Models

Reference: <https://cloud.google.com/vertex-ai/docs/quotas>

6.1.2 Production Quotas (After Increase Request)

- **US Regions** (us-central1): Up to 300-1000+ requests/minute
- **EU Regions** (europe-west1): Up to 100-300+ requests/minute
- **Online Inference:** Up to 30,000 requests/minute per region
- **Processing Time:** US regions process increases almost instantly; EU regions may take 1-2 days

6.2 Concurrent User Capacity

6.2.1 Calculation

Assuming:

- Average response time: 2-3 seconds per query
- Rate limit: 300 requests/minute (post-increase)

$$\begin{aligned}\text{Max concurrent queries} &= \frac{300 \text{ requests/min}}{60 \text{ sec/min}} \times 3 \text{ sec} \\ &= 15 \text{ simultaneous queries}\end{aligned}$$

However, with **multiple regions** and **auto-scaling backend**:

- **Single Region**: 15-20 concurrent users
- **Multi-Region** (US + EU): 30-40 concurrent users
- **With Context Caching**: Up to 2x improvement (30-80 concurrent users)

6.3 Auto-Scaling Architecture

The system is designed to scale automatically:

1. **Frontend**: Vercel auto-scales serverless functions (up to thousands of concurrent users)
2. **Backend**:
 - Cloud Run or Render auto-scales based on CPU/memory
 - Can handle 100-1000+ concurrent HTTP requests
3. **Gemini API**: Google-managed infrastructure scales automatically
4. **Bottleneck**: Gemini API rate limits (solvable with quota increases)

6.4 Scalability Best Practices

- **Request Quota Increases Early**: Submit requests 3-5 days before expected traffic surge
- **Use Multiple Regions**: Distribute load across US and EU regions
- **Implement Queueing**: For extremely high traffic, queue requests and process asynchronously
- **Monitor Usage**: Set up alerts in Google Cloud Console for approaching quota limits
- **Load Balancing**: Use Cloud Load Balancer for geographic distribution

6.5 Real-World Capacity Estimates

Note: These estimates assume quota increases have been approved and regional distribution is implemented.

Scenario	Concurrent Users	Queries/Month
Small Clinic	5-10	1,000-5,000
Medium Hospital	20-50	10,000-50,000
Large Health System	100-200	100,000-500,000
Regional Healthcare (multi-region)	500+	1,000,000+

Table 5: Capacity Estimates with Quota Increases

7 Additional Anticipated Questions

7.1 System Uptime & Reliability

Q: What is the expected uptime?

- **Google Cloud SLA:** 99.5-99.95% uptime for Vertex AI services
- **Gemini API:** Managed by Google with high availability
- **Frontend/Backend:** Depends on hosting provider (Vercel/Render offer 99.9% uptime)

Reference: <https://cloud.google.com/vertex-ai/sla>

7.2 Disaster Recovery & Backups

Q: How is data backed up?

- **Embeddings:** Google automatically replicates data across availability zones
- **Source Documents:** Recommend maintaining backups in Cloud Storage or on-premises
- **Configuration:** Backend code and settings versioned in Git
- **Recovery Time:** Gemini API has automatic failover; backend can be redeployed in minutes

7.3 Compliance & Regulations

Q: Does the system comply with Italian/EU healthcare regulations?

- **GDPR Compliant:** Yes, with data residency in EU regions
- **Data Processing Agreement (DPA):** Available from Google Cloud
- **HIPAA** (if applicable): Requires Business Associate Agreement (BAA) with Google
- **Encryption:** Meets European standards for data protection

Reference: <https://cloud.google.com/privacy/gdpr>

7.4 Integration Capabilities

Q: Can the chatbot integrate with existing systems?

- **REST API:** Fully exposed backend API for integration
- **SSO/SAML:** Can integrate with enterprise authentication systems
- **EHR Systems:** Possible to connect to electronic health records (requires development)
- **Webhooks:** Can send notifications to external systems
- **Embedding:** Chatbot widget can be embedded in existing websites

7.5 Monitoring & Logging

Q: How do we monitor system performance?

- **Google Cloud Monitoring:** Real-time dashboards for API usage, latency, errors
- **Cloud Logging:** Detailed logs for debugging and audit trails
- **Custom Metrics:** Track queries per domain, average response time, user satisfaction
- **Alerts:** Automated alerts for quota limits, errors, or unusual patterns

Reference: <https://cloud.google.com/monitoring>

7.6 Content Moderation & Safety

Q: How does the system prevent harmful outputs?

- **Built-in Safety Filters:** Gemini has filters for hate speech, dangerous content, etc.
- **Configurable Thresholds:** Can adjust safety settings (block none/few/some/most)
- **Grounding:** RAG ensures responses based on uploaded documents (reduces hallucinations)
- **System Instructions:** Explicitly instruct model to only answer from provided context

Reference: <https://cloud.google.com/vertex-ai/generative-ai/docs/multimodal/configure-safety-attributes>

7.7 Vendor Lock-in Considerations

Q: What if we want to switch providers in the future?

- **Data Portability:** Documents can be exported at any time
- **API Abstraction:** Backend architecture allows swapping AI providers
- **Open Standards:** Use standard vector embeddings (portable to other vector DBs)
- **Migration Path:** Can migrate to Anthropic Claude, OpenAI, or open-source models

8 Conclusion & Recommendations

8.1 Summary

The ULSS 9 Health Chatbot system is built on a **secure, scalable, and privacy-first architecture** using Google Cloud Platform:

- **Privacy:** Customer data is NOT used for public model training; enterprise controls available
- **RAG:** Gemini File Search provides managed, integrated RAG with minimal operational overhead
- **Costs:** Predictable pay-per-use pricing starting at \$0.33/month for 1,000 queries
- **Scalability:** Supports 15-200+ concurrent users (with quota increases and multi-region deployment)
- **Compliance:** GDPR, HIPAA, ISO 27001, and SOC 2/3 certified

8.2 Recommended Next Steps

1. **Request Quota Increase:** Submit request to Google Cloud for production quotas
2. **Enable VPC-SC & CMEK:** Implement enterprise security controls for maximum privacy
3. **Configure EU Residency:** Ensure data processing stays within European regions
4. **Sign DPA/BAA:** Formalize data processing agreements with Google Cloud
5. **Monitor & Optimize:** Set up comprehensive monitoring and implement context caching

8.3 Key Resources

All documentation and citations in this document are publicly accessible:

- **Google Cloud AI/ML Privacy:** <https://cloud.google.com/privacy>
- **Gemini API Documentation:** <https://ai.google.dev/gemini-api/docs>
- **Vertex AI Security:** <https://cloud.google.com/vertex-ai/docs/general/data-governance>
- **Pricing Calculator:** <https://ai.google.dev/pricing>
- **Compliance:** <https://cloud.google.com/security/compliance>